

Proposal in Response to RFQ – Solar-Powered Irrigation Systems

Submitted to: Alliance for a Green Revolution in Africa (AGRA)

Submitted by: SunHarvest Solutions Ltd.

Date: October 7, 2025

Page 1 – Executive Overview

SunHarvest Solutions Ltd. is honored to submit this response to AGRA’s September 2025 RFQ for the supply and installation of solar-powered irrigation systems in Kenya, Ghana, and Tanzania.

Our proposal is designed to deliver **reliable, affordable, and adaptable irrigation technologies** that align with AGRA’s mission of strengthening food security and smallholder farmer resilience.

Key features of our offer include:

- Provision of **solar-powered irrigation units** tailored to smallholder contexts.
- Training for farmer groups to ensure long-term operational capacity.
- Development of simplified training manuals, with partial translation into local languages.
- Commitment to monitoring and reporting, though with some limitations due to logistical constraints.

We acknowledge that certain aspects of our proposal will require refinement in collaboration with AGRA, particularly around **localized technical specifications and import cost assumptions**.

Page 2 – Technical Approach

System Components

Each solar irrigation package will typically consist of:

- A solar PV array (approx. 2–4kW depending on site).
- DC submersible pump (capacity: 5,000–7,500 liters/day).
- Storage tanks of 3,000–5,000 liters.
- Drip-line irrigation kit covering approximately 0.5–1.5 hectares.

Design Considerations

- Modular architecture allows scaling over time.
- Built-in protection against pump dry-running and electrical surges.
- Spare parts standardized to reduce downtime.

Gaps and Assumptions

- **Solar resource assessments** for each site are not yet included.
- **Water source feasibility** (depth, recharge rate) has not been independently verified.
- Remote monitoring options (via GSM) are considered optional, not standard.

While these details are important, we believe they can be resolved during the **implementation design phase**.

Page 3 – Deliverables & Training

Deliverables

1. Supply and installation of **9 solar irrigation systems** across three countries (3 per country).
2. Training for approximately **450 farmers**, covering equipment usage and maintenance.
3. Simplified manuals and pictorial guides; some translations planned for Swahili, though Ghanaian local language adaptations remain pending.
4. Initial after-sales support plan (12 months coverage).
5. Quarterly performance summaries, though in a simplified format.

Training Approach

- Practical demonstrations at each installation site.
- Use of farmer cooperatives to identify training participants.
- Incorporation of modules on **basic irrigation scheduling** and **water conservation techniques**.

Gap: The training does not yet cover all 500 farmers requested in the RFQ, and reporting formats may not fully align with AGRA's expectations.

Page 4 – Experience & Organizational Profile

SunHarvest Solutions Ltd. is a legally registered company operating in Kenya, with project partners in Ghana and Tanzania.

Relevant Experience

- **Kenya (2022–2023):** Installation of 6 pilot solar irrigation systems in Machakos County, funded by a bilateral donor.

- **Tanzania (2021):** Distribution of 20 portable solar pumps through a partnership with a farmer cooperative.
- **Ghana (2020):** Technical advisory support to an NGO-led irrigation project (indirect role).

Capacity

Our team includes engineers, agronomists, and field trainers with hands-on experience in solar pumping technologies. We also maintain partnerships with **regional suppliers for spare parts**.

Gap: We cannot yet provide three formal client references with verified contacts. One reference is ready, while the others are awaiting approval for disclosure.

Page 5 – Financial Proposal & Sustainability

Financial Proposal (USD, indicative)

Item	Unit Cost	Quantity	Subtotal
Solar irrigation unit (average package)	\$5,200	9	\$46,800
Farmer training (per session)	\$2,000	9	\$18,000
Spare parts (per system)	\$350	9	\$3,150
Manuals & materials	\$5,000	1	\$5,000
Monitoring & reporting (year 1)	\$7,500	1	\$7,500
Total (Excl. taxes & freight)			\$80,450

Gap: Excludes customs duties, inland transport, and VAT, which vary per country.

Sustainability Plan

- Local technicians to be identified and trained at each pilot site.
- Stocking of essential spare parts in Nairobi, with extension to Dar es Salaam.
- Proposal to establish farmer-led maintenance funds, though details are still under development.
- After-sales support limited to the first 12 months; longer-term arrangements negotiable.

Conclusion

SunHarvest Solutions is committed to partnering with AGRA to deliver scalable and environmentally sustainable irrigation solutions. While our current proposal does not fully address every element of the RFQ (notably the full farmer training numbers, localized

manuals, and long-term sustainability planning), we believe our **core offering provides a solid foundation** for collaboration and can be refined in dialogue with AGRA.

Submitted by:

SunHarvest Solutions Ltd.

Contact: info@sunharvest-africa.com